

Lab: Causal Reasoning and Problem Reframing

Workshop

Objective

Practice the core cognitive skills of inventive thinking: building causal models, reframing problems, running mental simulations, and comparing competing explanations. You will apply these skills to your own invention idea and discover how changing the question can open entirely new solution spaces.

Materials and Prerequisites

- Your invention idea from previous labs
- Paper and pen for diagramming
- A timer or clock
- No technical background required — all exercises use everyday reasoning

Part 1: Building a Causal Model of Your Problem (15 minutes)

Every invention addresses a problem, and every problem has a causal structure — a chain of causes and effects that produces the undesirable situation. Before you can solve the problem, you need to understand what causes it.

Step 1a: Symptom Identification

What is the observable symptom of the problem your invention addresses? Describe what a person would actually see, feel, or experience.

Write your response here

Step 1b: The Five Whys

Starting from your symptom, ask "why?" repeatedly to trace the causal chain backward to root causes. Record each level:

Level	Why?	Answer
Symptom	Why does [your symptom] happen?	

Write your response here

|| Cause 1 | Why does [your Cause 1] happen? |

Write your response here

|| Cause 2 | Why does [your Cause 2] happen? |

Write your response here

|| Cause 3 | Why does [your Cause 3] happen? |

Write your response here

| | Root Cause | Why does [your Cause 3] happen? |

Write your response here

|

Step 1c: Multiple Causes

Most problems have more than one causal pathway. Identify at least two additional causal chains that could produce the same symptom:

Alternative causal chain 1:

Write your response here

Alternative causal chain 2:

Write your response here

Step 1d: Causal Diagram

Draw a diagram connecting all the causes you have identified to the symptom. Use arrows to show the direction of causation. Include any feedback loops (where an effect also causes one of its causes).

Describe your causal diagram in words or upload a sketch:

Write your response here

Which cause do you think is the most important? Why?

Write your response here

Which cause would be easiest to intervene on? Is it the same as the most important cause?

Write your response here

Part 2: Problem Reframing Exercise (15 minutes)

Now practice changing the question. You will reframe your invention problem using three different techniques.

Technique 1: Inversion

Instead of "How do I solve this problem?" ask "How would I create this problem on purpose?"

If I wanted to deliberately cause the problem my invention addresses, what would I do?

Write your response here

What does this inversion reveal about the causal structure? Are there causes you had not considered?

Write your response here

Technique 2: Constraint Removal

List 5 constraints you are assuming about your invention (physical, economic, social, regulatory, or self-imposed):

#	Constraint	Is it real or self-imposed?
1		

Write your response here

Write your response here

|| 2 |

Write your response here

|

Write your response here

|| 3 |

Write your response here

|

Write your response here

|| 4 |

Write your response here

|

Write your response here

|| 5 |

Write your response here

|

Write your response here

|

Remove one self-imposed constraint. What new solutions become possible?

Write your response here

Technique 3: Abstraction

State your problem at three levels of abstraction:

Specific level (your exact problem):

Write your response here

Function level (the general function you are trying to achieve):

Write your response here

Value level (the human value you are trying to serve):

Write your response here

At the function level, what solutions exist that your specific-level framing would not have considered?

Write your response here

Which reframing technique produced the most surprising or useful new perspective?

Write your response here

Part 3: Mental Simulation of Candidate Solutions (15 minutes)

Choose three candidate solutions for your problem (your current best idea plus two alternatives) and run mental simulations.

Solution A (your current best idea):

Describe the solution:

Write your response here

Mental simulation — walk through how it works step by step:

Step	What happens	What could go wrong
1		

Write your response here

Write your response here

|| 2 |

Write your response here

|

Write your response here

|| 3 |

Write your response here

|

Write your response here

|| 4 |

Write your response here

|

Write your response here

|

Confidence that this solution would work (1-10):

Write your response here

Solution B (an alternative):

Describe the solution:

Write your response here

Mental simulation — walk through how it works step by step:

Step	What happens	What could go wrong
1		

Write your response here

|

Write your response here

|| 2 |

Write your response here

|

Write your response here

|| 3 |

Write your response here

|

Write your response here

|| 4 |

Write your response here

|

Write your response here

|

Confidence that this solution would work (1-10):

Write your response here

Solution C (a radically different approach):

Describe the solution:

Write your response here

Mental simulation — walk through how it works step by step:

Step	What happens	What could go wrong
1		

Write your response here

|

Write your response here

|| 2 |

Write your response here

|

Write your response here

|| 3 |

Write your response here

|

Write your response here

|| 4 |

Write your response here

|

Write your response here

|

Confidence that this solution would work (1-10):

Write your response here

Part 4: Counterfactual Reasoning (10 minutes)

For your most promising solution, answer these counterfactual questions:

If the materials were free, how would your solution change?

Write your response here

If the solution had to work for a child, how would it change?

Write your response here

If the solution had to work without electricity, how would it change?

Write your response here

If the solution had to be biodegradable, how would it change?

Write your response here

Which counterfactual scenario produced the most interesting design change?

Write your response here

Part 5: Cognitive Flexibility Self-Assessment (5 minutes)

Rate yourself honestly on these cognitive flexibility indicators:

Indicator	Rating (1-5)	Evidence from this lab
I generated multiple causal explanations for my problem		

Write your response here

|

Write your response here

| | I seriously considered solutions very different from my initial idea |

Write your response here

|

Write your response here

| | I changed my mind about something during this lab |

Write your response here

|

Write your response here

| | I identified assumptions I had not previously questioned |

Write your response here

|

Write your response here

| | I imagined how someone very different from me would approach this problem |

Write your response here

|

Write your response here

|

What is the biggest shift in your thinking that occurred during this lab?

Write your response here

Discussion and Debrief

1. **Causal depth:** How deep did your causal analysis go? Did the Five Whys reveal root causes that surprised you, or did you feel stuck at surface-level explanations?

2. **Reframing impact:** Which reframing technique (inversion, constraint removal, abstraction) was most generative for your specific problem? Why?
3. **Mental simulation limits:** Where did your mental simulation break down? What aspects of your solution could you not predict mentally? What does this tell you about what to test first when prototyping?
4. **Cognitive biases:** Did you notice any functional fixedness, Einstellung, or confirmation bias in your own thinking during this lab? What triggered it?
5. **Model competition:** After evaluating three candidate solutions, did your preference change from Solution A? What does this tell you about the value of generating alternatives before committing?

Write your response here